

Distinct Values Limit Collatz Correction: A Friendly Guide to the Sixth-Root Bound

Companion to “A Sixth-Root Bound on Collatz Correction”*

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1 An escaping orbit cannot reuse small numbers

Use the shortcut Collatz rule: halve an even number; triple an odd number, add one, and halve it. If the sequence ever visits the same number twice, its future repeats. The whole orbit is then bounded.

That means an unbounded orbit must keep visiting distinct numbers. In particular, it cannot repeatedly use the same small odd numbers to collect the effect of the $+1$ in the rule. This simple observation gives a quantitative bound.

Write N for the starting number, j for the number of odd steps taken so far, and c for the normalized affine correction used in the preceding reciprocal-sum paper. The normalization removes the powers of three and two from the accumulated additions.

The new Lean result. Every unbounded positive orbit obeys

$$(N + c)^6 \leq N^6(j + 1),$$

so c/N is at most $(j + 1)^{1/6} - 1$. No assumption that the reciprocal sum converges is needed.

For illustration, the bound allows $c/N \leq 1$ after 63 odd steps, $c/N \leq 2$ after 728 odd steps, and $c/N \leq 9$ after 999,999 odd steps. These are evaluations of the formula, not examples of unbounded Collatz orbits.

The theorem also works on a finite stretch beginning at the seed: if its visited values are distinct and avoid one, the bound holds through the end of that stretch. No assertion about its infinite future is required.

*Written with Codex (OpenAI). The technical paper and Lean source give the exact assumptions and proofs. No claim of literature priority is made.

2 Why a sixth root appears

At an odd value x , the quantity $N + c$ is multiplied by

$$1 + \frac{1}{3x}.$$

At an even value it is unchanged. For every $x > 1$, Lean checks the polynomial inequality

$$\left(1 + \frac{1}{3x}\right)^6 \leq \frac{x+1}{x-1}.$$

For an odd value $x = 2k + 1$ greater than one, the right side becomes $(k + 1)/k$. The odd values are distinct, so the positive integers k are distinct too.

The largest product of j factors of the form $(k + 1)/k$ comes from the smallest possible distinct positive k values. Those factors cancel:

$$\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{j+1}{j} = j + 1.$$

This bounds the sixth power of the total correction factor by $j + 1$. Taking a sixth root gives the announced estimate. The proof uses distinctness and exact arithmetic, not a random-parity model.

3 What it contributes to the larger proof

The earlier result said that bounded correction is equivalent to a finite reciprocal orbit sum. It left open the possibility of an escaping orbit whose reciprocal sum diverges. The new theorem applies to that case as well: even if its correction is unbounded, the correction must grow within this slow envelope.

There is also a constraint on odd steps. After L steps, if an unbounded orbit has not dropped below its starting value, then

$$2^{6L} \leq 3^{6j}(j + 1).$$

This places the odd-step count close to or above the familiar critical line, allowing only a logarithmic deficit. Lean proves the displayed integer inequality directly.

What is still missing. Slow growth is still growth. The sixth-root bound does not make correction bounded, and it does not force descent. The next page uses this estimate to exclude slow multiplicative drift. We still need to exclude the remaining itineraries. Nontrivial cycles remain a separate question.

The implementation is `lean/Collatz/CorrectionGrowth.lean`. The technical paper contains the complete proof and rebuild instructions. The expanded audit covers 57 selected declarations with only standard foundational axioms. Collatz has not been proved.

4 A consequence: slow drift cannot escape

The correction estimate now gives an exclusion theorem. Let

$$M_t = \frac{3^{j_t}}{2^t},$$

where j_t counts odd steps before time t . This ratio measures the multiplicative part of the orbit rule; it leaves out the additions tracked by c_t .

Among the first $L + 1$ values of an unbounded orbit, at least one must be at least $L + 1$. Otherwise there would not be enough distinct positive integers to fill the list. Combining that fact with the sixth-root correction estimate shows that at some time $t \leq L$,

$$M_t \geq \frac{(L + 1)^{5/6}}{N}.$$

This holds somewhere in each prefix. It does not say that the ratio is this large at every time.

New exclusion. If the multiplicative ratio stays below a polynomial bound with rational exponent strictly less than $5/6$, the orbit is bounded. In particular, a uniformly bounded multiplicative ratio cannot drive an unbounded orbit.

For example, a bound $M_t \leq A(t + 1)^{2/3}$, for a fixed $A > 0$ and every time t , is too slow to support an unbounded orbit. Such a bound is an explicit hypothesis, not something proved for arbitrary Collatz seeds.

Only a finite stretch needs checking

There is also an exact integer version. Choose nonnegative integers a, b, C with $b < 5a$. If

$$M_t^{6a} \leq C(t + 1)^b$$

holds at every time through $L = CN^{6a}$, the orbit is bounded. Powers of two and three can replace the fractions, so the test requires no floating-point arithmetic. The sufficient horizon can be very large.

The remaining question

An unbounded orbit would have to exceed every such slow envelope. The theorem does not exclude faster drift, and a bounded orbit has not thereby been proved to reach one. Those are still mathematical gaps, not verification tasks. The new source is `lean/Collatz/DriftEscape.lean`; its four main results are included in the 57-declaration audit.